TCU 版 (vortex_llm) PR 各阶段时序路径报告

设计: Vortex (GPU, TCU 开) | 工艺: SKY130 | 目标: 50MHz (20ns) | tt/25C/1.8V

Floorplan (2_floorplan_final)

指标	值
Worst Slack (setup)	-33.69 ns
TNS	-397156.47 ns
fmax	53.69ns → 18.6MHz
Setup Skew	-

关键路径明细 (setup)

```
Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.issue.g_slices[0].issue_slice
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
```

Global Place (3_global_place)

指标	值
Worst Slack (setup)	2.32 ns
TNS	0.00 ns
fmax	17.68ns → 56.6MHz
Setup Skew	-

关键路径明细 (setup)

```
Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0].
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
```

Resizer (3_resizer)

指标	值
Worst Slack (setup)	2.32 ns
TNS	0.00 ns
fmax	17.68ns → 56.6MHz
Setup Skew	–

关键路径明细 (setup)

```
Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0].
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
```

Detailed Place (3_detailed_place)

指标	值
Worst Slack (setup)	2.62 ns
TNS	0.00 ns
fmax	17.38ns → 57.5MHz
Setup Skew	–

关键路径明细 (setup)

```
Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0].
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
```

CTS (4_cts_final)

指标	值
Worst Slack (setup)	2.50 ns
TNS	0.00 ns

指标	值
fmax	17.50ns → 57.1MHz
Setup Skew	0.53 ns

关键路径明细 (setup)

```
Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0].
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap     Slew   Delay    Time   Description
```

Global Route (5_global_route)

指标	值
Worst Slack (setup)	1.09 ns
TNS	0.00 ns
fmax	18.91ns → 52.9MHz
Setup Skew	0.53 ns

关键路径明细 (setup)

```
Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0].
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap     Slew   Delay    Time   Description
```

Signoff(RCX) (6_finish)

指标	值
Worst Slack (setup)	2.29 ns
TNS	0.00 ns
fmax	17.71ns → 56.5MHz
Setup Skew	0.55 ns

关键路径明细 (setup)

```
 Startpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0
             (rising edge-triggered flip-flop clocked by clk)
Endpoint: g_clusters[0].cluster.g_sockets[0].socket.g_cores[0].core.execute.fpu_unit.g_blocks[0].
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
```